Hospital Appointment Platform for India: Research & Feature Recommendations

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Executive summary

Winning in India depends on seven decisions: trust in the booking, phone-first access, and staying inside the new advertising rules. Most incumbents lose users on refunds, family mix-ups and slots that turn out not to exist, so those are our openings.

1. **Log in with a phone number and a one-time password (OTP).** Keep email as a secondary field. Send every booking update on WhatsApp and SMS, with in-app notifications as a backup.
2. **Make the booking flow short and configurable.** Each hospital chooses, per doctor, between auto-confirm and manual approval. Manual approval gets a time limit, after which the booking is auto-cancelled with a refund. This replaces the three-step chain of user, then doctor, then hospital.
3. **Sell promotion to hospitals, never to doctors, and label it "Sponsored".** NMC guidelines issued on 6 October 2026 prohibit paid ranking of doctors on platforms (source). Sponsored hospitals get a capped boost on top of a ranking that is otherwise organic.
4. **Connect to ABDM early.** India has 90 crore ABHA (health ID) accounts (source). ABHA Scan & Share has issued over 2 crore OPD tokens at 3,860+ facilities (source). Linking a patient's ABHA and the hospital's facility registry entry is a trust signal and saves patients typing.
5. **Make pay-at-hospital the default, with UPI prepayment as an option.** Every prepaid booking needs an automatic refund path. Practo, NH Care and Lybrate reviews show failed refunds destroy trust (Practo listing).
6. **Give every family member a separate profile.** Each person gets their own token and hospital ID. Manipal and Eka Care reviews show tokens and profiles merging into the account holder's name.
7. **Build DPDP compliance into the first version.** That means consent notices, data deletion and breach logging. The main duties under the DPDP Rules apply from 13 May 2027 (source).

The MVP should do three things well for one city: hospital search by distance, real slots, and a confirmation on WhatsApp. Hindi, token queues and ABDM follow in Phase 2.

Competitive teardown

No Indian app does multi-hospital booking with reliable slots, clean family profiles and honest sponsored ranking at the same time. That gap is our position. Ratings and download counts are from Google Play listings, checked on 8 November 2026. The weaknesses come from the reviews shown on each listing.

App	Type	Standout feature	Weakness/gap	What we should copy
Practo	Aggregator plus clinic and hospital software	4.5★, 1 crore+ downloads. 1 lakh+ doctors and 9,000+ bookable hospitals and clinics (about). Free follow-up chat after a visit	Paid bookings that fail get no refund or reschedule. Health-checkup slots run late. Support is hard to reach in the app	Hospital search by criteria. A free follow-up after the visit. Automatic refund when a booking breaks
Apollo 24|7	Hospital chain, now pharmacy-led	5,000+ doctors, 70 hospitals, 4,000+ pharmacies. Doctors, labs and pharmacy in one app	No cash on delivery. Orders can't be cancelled for a refund. Cashback not credited	Exact fees and refund terms shown before payment
Max MyHealth	Hospital-chain app	Family can join video consults. Live inpatient bill updates. Duplicate patient profiles can be merged. One-tap ambulance or route to the nearest hospital	Video bugs. No field for a corporate tie-up discount	Merging duplicate profiles. A nearest-hospital and ambulance button

App	Type	Standout feature	Weakness/gap	What we should copy
Fortis (myFortis)	Hospital-chain app	10 lakh+ downloads. Books for family members and takes online payment	Doctor turns out to be in surgery after the slot was confirmed. Some services can't be booked. No invoices or reports in the app	Real-time alert when a doctor becomes unavailable. Invoices and reports in the app
Manipal Hospitals	Hospital-chain app	Bill payment in the app. Health packages and home care	Slow app. An update overwrote family members' profiles with the account holder's name. Past visits listed above upcoming ones	Strictly separate family profiles. Upcoming appointments listed first
Narayana Health (NH Care)	Hospital-chain app	4.8★. Same-day slots. Pay online to skip the queue. Reports can be shared with caregivers	Hidden ₹1,000 home-collection fee. Lab bookings can't be cancelled. No way to save a favourite branch	Same-day slots. Pay online to skip the queue. A saved "my hospital"
MediBuddy	Aggregator focused on corporate and OPD insurance	Cashless booking, network-hospital locator, claim tracking. Links Ayushman (PM-JAY) and ABHA cards	Support is a bot only. Prescriptions missing after consults. Prices not shown upfront	A "Cashless here" badge and an insurance-network filter

App	Type	Standout feature	Weakness/gap	What we should copy
Tata 1mg	Pharmacy-led	1 crore+ downloads. Medicines delivered in 1,000+ cities. NABL- and CAP-accredited labs	Up to 6 sales calls a day about abandoned carts	Accreditation badges as trust signals. Users can opt out of follow-up calls
Eka Care	Health records and ABHA app, also runs OPD token booking	1 crore+ downloads. Creates ABHA in the app. Family health records	OPD token issued in the account holder's name, not the patient's. Accounts can't be deleted	ABHA creation during sign-up. Tokens issued per patient. Self-service account deletion
Lybrate	Aggregator for online consults	Free anonymous health questions answered by doctors	3.1★, last updated June 2023. Doctors don't call for paid consults	A warning: trust collapses without no-show refunds and support
Bajaj Finserv Health	Aggregator offering health plans and EMI	Claims 5,500+ hospitals, 2,500+ labs and 1 lakh+ doctors. Bundled OPD plans	Plan services can't be booked. Claim uploads fail	EMI or plan badges on hospital pages, in Phase 3
eSanjeevani	Government telemedicine	50 crore+ teleconsults, 15 languages (source)	2.5★. Only one family member per phone number. Doctor hours not shown	Many languages. Several family members on one number

App	Type	Standout feature	Weakness/gap	What we should copy
ABHA app (NHA)	Government health-records app	1 crore+ downloads. QR Scan & Share at the hospital counter. Consent-based record sharing that users can revoke	Token holders still queue 2–3 hours because there is no separate line	Scan & Share at the desk, paired with a real queue position on the hospital side
Zocdoc (US)	Global reference	Real-time availability. Reviews only from verified patients. Filter by insurance accepted	Sponsored profiles repeat as users scroll	Live slots and verified reviews. Show each sponsored hospital once and cap it
NHS App (UK)	Global reference (government)	Switch profiles to act for a child or family member, with permission. Hospital messages inbox. Test results in the app	Login lockouts after too many code requests	Proxy access for family members. A simple login that never locks users out

Patterns and UX lessons

The best apps make booking feel certain. Users can book in three taps, the slot is real, and the price is final. The worst reviews come when any one of those breaks. Each pattern below is drawn from the teardown above.

- **Book in three taps or fewer.** The steps are hospital, then doctor or department, then a slot from a date strip and time chips. Ask for name and age only at confirmation. NH Care (4.8★) and Zocdoc put the slot picker on the first screen.
- **Show only slots that are real.** Grey out taken slots and block a slot while someone is booking it. When a doctor becomes unavailable, notify patients at once and offer the next free slot. Fortis reviews show the cost of getting this wrong.

- **Show the full price before the user commits.** Show the consultation fee, any convenience fee and the refund rule on the slot screen. NH Care's undisclosed ₹1,000 home-collection fee and MediBuddy's missing prices are among the top complaints.
- **Make cancel and reschedule one tap each.** Refunds should be automatic, with the timeline shown, for example "UPI refund in 3–5 working days" (approximate). Practo, Apollo and Lybrate lose ratings over manual refunds.
- **Keep each family member's profile separate.** Each member has their own name, age, ABHA and hospital ID. The booking screen should ask "Who is this for?" with a picker. Manipal, Eka Care and eSanjeevani reviews show mix-ups.
- **Let users save their hospital.** "My hospital" and "My doctor" sit on the home screen for one-tap rebooking. NH Care users ask for this.
- **Use trust badges built from facts.** Examples: NABH or NABL accreditation, "ABDM registered", "Cashless here", and "Verified patient" reviews only. Superlatives like "Best" or "No.1" are not allowed under NMC guidelines.
- **Mark sponsored listings, show each once, and cap them.** Zocdoc reviews complain that sponsored profiles repeat as users scroll.
- **Make sure users can reach a person.** Support should offer a call button or a WhatsApp chat with the hospital desk. Almost every Indian app reviewed fails here.
- **Don't lock users out at login.** Allow an OTP resend after 30 seconds and a voice OTP as a fallback, with no long lockouts. NHS App reviews cite 15-minute lockouts.
- **For the hospital staff side, design for a non-technical front desk.** Use big buttons: Accept, Reject, Reschedule, Mark arrived, Mark no-show. Show a single "Today" queue view and use no free-text forms in the main path.

India-specific success factors

In India the booking product is only half the work. The other half is meeting patients where they already are: a phone number, WhatsApp, UPI, their own language, and an OPD queue at the hospital. The table ranks each factor by how much it moves adoption.

Factor	What the evidence says	What we build	Phase
Phone OTP login	Patients identify by phone number. eSanjeevani's limit of one family member per number is a top complaint (listing). SMS needs TRAI DLT registration of the sender ID and every template (MSG91)	Phone OTP as the primary login, with email optional. Start DLT registration in week 1 because approval takes days. Username and password come later, as the brief says	MVP

Factor	What the evidence says	What we build	Phase
WhatsApp notifications	Utility messages cost ₹0.115 each, before 18% GST. They are free inside the 24-hour window after a user messages us. Marketing messages cost ₹0.86 (rates, Oct 2026)	Confirmation, reminder 2 hours before, and token-is-near alerts sent as utility templates. SMS as the fallback	MVP
UPI payments	25.47 billion UPI transactions worth ₹31.53 lakh crore in October 2026 (source)	UPI intent or QR through Razorpay for optional prepayment. Automatic refund on cancellation	Phase 2
Cash at the counter	Hospitals expect patients to pay at the desk (approximate, based on how India's OPDs work; to confirm with the client)	"Pay at hospital" as the default. The front desk marks it Paid with one button	MVP
Language: Hindi and regional	57% of internet users prefer regional-language content, and 488 million of 885 million users are rural (Kantar/IAMAI 2024). eSanjeevani runs in 15 languages	Hindi and English at launch, using next-intl in Next.js. Add the hospital's state language next. Use icons plus short words on every button	Phase 2
Family profiles	Profile and token mix-ups appear in Manipal, Eka Care and eSanjeevani reviews	Up to 6 members under one phone number (approximate cap), each with their own name, age, gender and ABHA	MVP
ABDM and ABHA	90 crore ABHAs (source). Scan & Share cut OPD registration waits from 50 to 4 minutes at about 347 hospitals (source)	Optional ABHA field on each profile. Show an HFR-verified badge on hospital pages. Full Scan & Share integration later	Phase 2–3
OPD tokens and queues	Indian OPDs run on token numbers. ABHA app users still wait 2–3 hours when tokens aren't tied to a real queue (listing)	Each confirmed booking gets a token number per doctor per day. A live "Now serving #12" view, with the hospital pressing Next	Phase 2

Factor	What the evidence says	What we build	Phase
Walk-ins	Government OPDs still see heavy walk-in traffic. At PGI Chandigarh in September 2021, 3,942 people walked in on the day 1,345 registered by teleconsultation (source)	The front desk adds a walk-in in 2 taps (name and phone). It takes the next token, and online slots and walk-ins share one queue	MVP (desk entry)
Low bandwidth	Users average 90 minutes online a day, and many are on budget Android phones (approximate)	Pages under 200 KB, server-rendered lists, compressed images, no autoplay video. Bookings survive a dropped connection	MVP
Tier-2 and tier-3 cities	Tier-2 OPD consultations grew 83% year-on-year against 73% in metros (Pristyn Care release, Feb 2024)	Launch in one tier-2 city where the client has hospital relationships. Search by PIN code and locality, not only by GPS	MVP

Feature roadmap by role

The MVP is about one promise: a patient finds a nearby hospital, books a real slot, and gets a confirmation on WhatsApp. Everything else waits until hospitals use the system daily. Rows are grouped by phase, then by role.

Feature	Role	Phase	Why
Phone OTP login, with email as a secondary field	User	MVP	Phone is how Indian patients identify. Email-only login loses users
Nearest hospitals with distance, plus a PIN code search	User	MVP	The core discovery need in the brief. PIN code search covers users who deny GPS
Hospital page: departments, doctors, timings, fees, events and activities	User	MVP	The brief asks for events. Factual pages also stay within the NMC guidelines
Slot picker: date strip and time chips, booking in 3 taps	User	MVP	Slot friction is the main reason users drop off

Feature	Role	Phase	Why
Family profiles, with a "Who is this for?" picker	User	MVP	Mix-ups are a top complaint about competitors
My bookings: cancel and reschedule with one tap each	User	MVP	Refund and cancel failures sink incumbents' ratings
WhatsApp and SMS confirmation and a reminder	User	MVP	The highest-reach channel in India
Doctor and slot setup: weekly schedule, slot length, holidays	Hospital	MVP	No slots, no product. Must be doable by non-technical staff
Approval mode per doctor (auto or manual) plus an Accept/Reject inbox	Hospital	MVP	Replaces the three-step chain. See the booking flow
Today's list: Arrived, No-show and Paid buttons, plus walk-in entry	Hospital	MVP	Front-desk adoption decides whether the platform is used daily
Event and activity posts (camps, talks, vaccination drives)	Hospital	MVP	Asked for in the brief. A free reason for patients to come back
Hospital onboarding and verification (registration number, documents)	Admin	MVP	Fake listings kill trust. Collect the HFR ID when the hospital has one
Sponsored placement: on/off per hospital, start and end dates, a cap	Admin	MVP	The brief's revenue line. Must be labelled from day 1
Basic dashboard: bookings, cancellations and no-shows per hospital	Admin	MVP	Shows value to hospitals and drives renewals
Hindi interface	User	Phase 2	57% of internet users prefer regional languages
UPI prepayment with automatic refunds (Razorpay)	User	Phase 2	Cuts no-shows. Add after pay-at-hospital is proven
Live token queue: "Now serving #12" and an estimated wait	User + Hospital	Phase 2	Fits how Indian OPDs work. Fixes the gap seen in ABHA tokens

Feature	Role	Phase	Why
Verified reviews, only after a visit marked Arrived	User	Phase 2	Builds trust without fake reviews, which the NMC guidelines ban
Link an ABHA to a profile, and an ABDM-verified hospital badge	User + Hospital	Phase 2	90 crore ABHAs. A government-backed trust signal
Staff accounts with roles (reception, doctor, manager)	Hospital	Phase 2	Doctors approve on their phone and the desk handles the rest
Promotion analytics: impressions, clicks, bookings from sponsored placements	Admin + Hospital	Phase 2	Justifies the promotion price
Doctor app view: today's patients, approve and reschedule	Hospital (doctor)	Phase 2	Only needed where manual approval stays on
Username and password login	User	Phase 2	In the brief. Use as a fallback to OTP
Regional languages (Marathi, Tamil, Telugu, Bengali, and others)	User	Phase 3	Expand in the order of the cities we launch in
ABDM Scan & Share and health-record sharing with consent	User + Hospital	Phase 3	Needs certification in the ABDM sandbox first
Teleconsultation	User + Hospital	Phase 3	Brings in the Telemedicine Practice Guidelines
Lab test and pharmacy partner bookings	User	Phase 3	A second revenue stream
Invoices and reports in the app	User + Hospital	Phase 3	Fortis and MediBuddy users ask for this
Subscription billing per hospital, with GST invoices	Admin	Phase 3	Needed once there are more than about 20 paying hospitals (approximate)

Promotions and recommendations within Indian rules

Promote hospitals, never individual doctors. Label every paid placement "Sponsored" and cap how far money can move a result. The rules tightened on 6 October 2026: NMC's Guidelines on Ethical Advertising and Public Communication say "all healthcare platforms hosting RMP shall ensure compliance" and they prohibit "paid ranking of doctors" (source).

What the rules say

Rule	What it covers	Impact on our "Top recommendations"
NMC Guidelines on Ethical Advertising (6 Oct 2026)	Applies to doctors and hospitals. It bans paid ranking of doctors, fake reviews, manipulated search rankings, unverifiable superlatives and discounts used to induce patients. Doctors' photos may not be used for self-promotion. Hospitals may publish factual information (departments, facilities, fees, accreditation). A doctor directory on a hospital site counts as information, not promotion. Penalties for doctors run from a warning to removal from the register for up to 3 years	Sell sponsored placement to hospitals only. Never sell doctor ranking. No "Best" or "No.1" badges. Reviews only from verified visits
NMC Professional Conduct Regulations 2023	Notified in 2023, then held in abeyance by a 23 August 2023 notification. The 2002 IMC code applies meanwhile, and it bars doctors from self-promotion (source)	Doctors cannot pay us to be featured

Rule	What it covers	Impact on our "Top recommendations"
ASCI Code and healthcare complaints	Healthcare was 21% of ASCI complaints in April–September 2023, the most of any sector. ASCI decisions require proof for "No.1" or "most trusted" claims, and its awards guideline requires the awarding body and date to be named. Not verified: ascionline.in did not load	Hospital ad copy goes through a checklist in the admin panel. No superlatives or outcome claims
Drugs and Magic Remedies Act, 1954	Bans ads claiming to cure the 54 scheduled conditions, which include cancer, diabetes and heart disease. Up to 6 months' prison for a first offence	Block "cures diabetes" or "guaranteed" wording in event and promotion text with a keyword filter plus admin review
CCPA Dark Patterns Guidelines, 30 Nov 2023	Bans 13 patterns, including disguised advertisement (ads blended to look like organic content), drip pricing and false urgency	A "Sponsored" label on every card. Show fees upfront. No "Only 1 slot left!" unless it is true

Recommended ranking rule

1. **Filter first.** Only show hospitals that are verified, open, and have a free slot in the next 7 days for the searched specialty.
2. **Rank organically.** Score each hospital on distance (50%), verified rating (25%) and soonest available slot (25%). The weights are a starting point to tune.
3. **Apply a capped sponsored boost.** A sponsored hospital can move up at most 3 places, and only inside a 10 km radius. At most 2 sponsored cards appear in the top 5, each shown once per page.
4. **Label clearly.** Use a visible "Sponsored" tag on the card itself, not hidden in an info icon. Add a "Why am I seeing this?" link.
5. **Keep a record.** Store every promotion contract with its dates and placement in the admin panel, as an audit trail if a complaint arrives.

Compliance and data

Build for the DPDP Act from the first release. The main duties apply from 13 May 2027, about six months away, and retrofitting consent and deletion later costs more than building them in now. Health data is the most sensitive data we will hold, so we keep it in India, minimise it, and log every access.

Rule	Status as of 8 Nov 2026	What we must build
DPDP Act 2023 and DPDP Rules 2025 (G.S.R. 846(E), 13 Nov 2025)	Data Protection Board rules in force. Consent-manager rule from 13 Nov 2026. Notice, consent, security and breach duties from 13 May 2027. A January 2026 proposal to cut 18 months to 12 was not notified as of September 2026 (source)	A standalone plain-language consent notice at signup, available in Hindi on request. A consent log. "Download my data" and "Delete my account" buttons. Verifiable parent consent for patients under 18. Breach notice to the Board with details within 72 hours. Penalties go up to ₹250 crore for security failures
Telemedicine Practice Guidelines (25 Mar 2020)	In force. Only applies if we add teleconsultation (Phase 3)	Verify each doctor's council registration before listing. Display name, qualification and registration number. Run grievance redressal. No AI prescribing or counselling
ABDM registries: HFR, HPR and sandbox milestones	Voluntary for private platforms, but a strong trust signal	MVP: store each hospital's HFR ID and each doctor's HPR or council ID, checked by an admin. Phase 3: pass sandbox milestone M1 (ABHA creation and verification), then M2 and M3 if we hold records. Certification follows NHA testing

Rule	Status as of 8 Nov 2026	What we must build
CERT-In directions (28 Apr 2022)	In force	Report cyber incidents within 6 hours. Keep system logs for 180 days within India
NMC Ethical Advertising Guidelines (6 Oct 2026)	In force	See the Promotions section. Patient photos or testimonials only with documented consent

Data residency and handling

- **Host in an Indian cloud region**, such as AWS Mumbai or Hyderabad, or GCP Mumbai or Delhi, for the database, backups and logs. DPDP allows transfers abroad unless the government restricts a country. Hosting in India still keeps us clear of future restrictions and of CERT-In's log rule.
- **Collect the minimum.** At booking that means name, age, gender, phone and reason for visit (optional, short). No diagnosis or report uploads in the MVP.
- **Limit access by role.** A hospital sees only its own patients. Admin access to patient data is logged.
- **Encrypt data** in transit (TLS) and at rest. Mask phone numbers in admin lists.
- **Retain data for a fixed period.** Keep bookings and audit logs for at least 1 year (Rules 6 and 8). Erase inactive accounts after a 48-hour warning, as the Rules' Third Schedule describes.
- **Separate the paperwork.** We are a data fiduciary for patient accounts. Each hospital should sign a data processing agreement covering the data it sees.

Monetization models

Start with a flat monthly subscription per hospital plus sponsored hospital placement. Add a small per-booking fee only once hospitals see bookings arriving. Practo's history shows clinic software subscriptions can carry an Indian healthtech business. A 2021 analysis called Ray its "largest source of revenue" (source). Rows run from what we should launch with to what we should add last.

Model	Who pays	Indian reference point	Fit for us	Phase
SaaS subscription per hospital	Hospital	Practo Ray was ₹999 and ₹1,999 a month per clinic in 2015 (source). Current price not published	High. Predictable, and it doesn't touch the doctor-patient relationship. Suggest ₹1,500–5,000 a month by hospital size (approximate, to test)	MVP: free pilot for 3 months, then paid
Sponsored hospital placement	Hospital	Practo sold paid doctor placement as "Practo Profile" (source). Since 6 Oct 2026, NMC bans paid ranking of doctors	Medium-high, if it is hospital-level only, labelled, capped and sold by city and period	MVP
Per-booking fee	Hospital	Practo started with a percentage of each booking (source)	Medium. Charge only for bookings marked Arrived, so no-shows are not billed. Avoid fees that look like a patient referral commission, which the NMC guidelines prohibit	Phase 2
Convenience fee	Patient	Razorpay charges 2% plus GST per transaction (pricing). Platforms often pass this to the user	Low early on. It adds friction, and CCPA's drip-pricing rule means it must be shown upfront. Only for prepaid slots	Phase 2
Lab and pharmacy partnerships	Partner	Tata 1mg and Apollo 24|7 run their own labs and pharmacies (see the teardown)	Medium, once there is traffic. Commission on bookings at partner labs	Phase 3

Model	Who pays	Indian reference point	Fit for us	Phase
Patient membership	Patient	Practo Plus offers unlimited online consults for a yearly fee. Apollo has Circle	Low for a booking-first product. Revisit with teleconsultation	Phase 3
Hospital software add-ons (queue screen, SMS packs, analytics)	Hospital	Practo Ray's higher tier added an IVR and missed-call alerts (source)	Medium. A TV queue screen and extra WhatsApp volume are easy upsells	Phase 2–3

One pricing rule protects all of these: never charge a hospital per patient sent its way in a form that reads as a referral fee or commission. The NMC guidelines bar "referral fees, commissions, or agency payments linked to patient procurement". A flat fee or a fee for software usage is safer. This needs a check by counsel.

Technical implementation notes

The planned stack of Next.js, Express, Prisma and PostgreSQL handles everything in the MVP. Two places need care: making double-booking impossible at the database level, and sending notifications through a queue rather than inside the request. The table maps each concern to the stack.

Concern	Recommendation	Stack detail
Nearest-hospital search	MVP: a Haversine distance in SQL over `lat` / `lng` columns, pre-filtered by a bounding box (±0.1° is about 11 km) so the index is used. Later: PostGIS `geography` with a GiST index and `ST_DWithin`	Prisma has no PostGIS type. Use `Unsupported("geography")` plus `$queryRaw` for the search. SQLite in dev can't run PostGIS, so keep the Haversine version as the dev path or move dev to Postgres in Docker early
Slots	Generate slots from each doctor's weekly schedule, slot length and holidays. Store booked slots as rows, not free ones	One `Appointment` row per booking, with `doctorId`, `slotStart` and `status`

Concern	Recommendation	Stack detail
Stopping double-booking	A unique index on (`doctorId`, `slotStart`) for active statuses. Create the booking in a transaction and catch the unique-violation error (`P2002`) as "slot just taken". Optionally hold a slot for 5 minutes during payment with a `HELD` status and an `expiresAt`	A Prisma `@@unique` cannot be partial. Add a raw SQL migration for `WHERE status IN ('HELD','PENDING','CONFIRMED')`. Test with two parallel requests
Status changes	A single `transition(appointment, event)` function that checks allowed moves and writes an `AppointmentEvent` audit row	Prevents illegal states such as a cancelled booking later confirmed. The audit row also feeds the dashboard
SLA timeout	A worker checks `PENDING` bookings past `approveBy` and auto-cancels them, refunding if prepaid	BullMQ delayed jobs with Redis, or node-cron every minute for the MVP
OTP login	MSG91 OTP API with a DLT-registered sender ID and template (DLT). Rate-limit to 3 OTPs per 10 minutes per number. Offer a voice OTP fallback	On verify, issue a short-lived JWT access token (15 minutes) and a rotating refresh token in an httpOnly cookie
WhatsApp	WhatsApp Cloud API directly, or a provider such as Gupshup, Interakt or MSG91. Get utility templates pre-approved: booking received, confirmed, reminder, cancelled, token near	Send from the queue, never inside the API request. Fall back to SMS if delivery fails
Payments (Phase 2)	Razorpay Orders and Checkout with UPI intent, or Payment Links. Confirm only on the signed `payment.captured` webhook. Refunds through the API	Pricing is 2% plus GST per transaction (pricing). Store the `razorpay_payment_id` and use idempotency keys

Concern	Recommendation	Stack detail
Notifications	One `notify(user, event)` service that writes to an outbox table. A worker sends WhatsApp, SMS and in-app messages, with retries	The hospital inbox uses polling every 15 seconds for the MVP. Add Server-Sent Events later
Roles	User, Hospital staff (reception, doctor, manager) and Admin. Every hospital query is scoped by `hospitalId` from the JWT	Express middleware `requireRole()` plus a `scopeToHospital()` helper
Language	next-intl with `/hi` and `/en` routes. All strings in JSON from day 1, even before the Hindi translation exists	Server-render hospital lists so pages load fast on 3G
ABDM (Phase 3)	Register on the ABDM sandbox and build milestone M1 (ABHA creation and verification) first (docs)	Keep the ABHA address and number as optional fields on `PatientProfile` now, so no migration is needed later
Hosting	Next.js on Vercel (Mumbai region) or AWS. Express and Postgres in AWS Mumbai (`ap-south-1`) on RDS with daily backups	Matches the data-residency guidance in the Compliance section

Keep the data model small. The MVP needs these tables: `User`, `PatientProfile` (family members), `Hospital`, `Department`, `Doctor`, `Schedule`, `Appointment`, `AppointmentEvent`, `Event` (hospital activities), `Promotion`, `Review`, `Notification`.

Risks, open questions and sources

The biggest risk is not technical. It is whether hospital front desks keep slots up to date every day. The second is the new NMC ban on paid ranking, which constrains how "top recommendations" can be sold.

Risks

Risk	Likelihood	Mitigation
Hospitals don't update slots or approve in time, so patients arrive to a closed OPD	High	Auto-confirm by default. SLA auto-cancel. A one-tap "Doctor on leave today" button that notifies all booked patients
Sponsored ranking challenged under the NMC guidelines (6 Oct 2026) or CCPA dark-patterns rules	Medium	Hospital-level only, labelled, capped, contracts logged. Legal review before launch
No-shows on free bookings	High (no reliable Indian figure found)	WhatsApp reminder 2 hours before. Optional UPI prepay in Phase 2. Allow controlled overbooking per doctor
DPDP obligations start 13 May 2027, possibly earlier if MeitY shortens the timeline	Medium	Build the consent, deletion and breach log in the MVP
WhatsApp or SMS price changes, or template rejection	Medium	Utility templates only. SMS fallback. Budget per booking tracked in admin
Fake hospital or doctor listings	Medium	Admin verification with registration number and HFR or council ID before going live
Incumbents (Practo, Eka Care) sign the same hospitals	Medium	Win on the hospital desk experience, local language and a city-by-city focus

Open questions for the client

- [ ] Which city and which hospitals launch first? How many have agreed to a pilot?
- [ ] Private hospitals only, or government hospitals too? Government OPDs already use ABHA tokens.
- [ ] Must the doctor personally approve each booking, or is approval by the hospital desk enough? We recommend auto-confirm with a manual option per doctor.
- [ ] Is booking free for patients? Will we take payment, or is it pay-at-hospital only?
- [ ] What will a hospital pay: a monthly fee, a promotion fee, or both? What price range?

- [ ] Should promotions ever cover individual doctors? We recommend no, because of the 6 Oct 2026 NMC rule.
- [ ] Who checks hospital event posts before they go live, the admin or the hospital itself?
- [ ] Which languages first: Hindi only, or a state language too?
- [ ] Is teleconsultation in scope at any point? It brings in the Telemedicine Practice Guidelines.
- [ ] Why add a username and password later? Is phone OTP plus email enough?
- [ ] Should patients be able to leave reviews in the MVP?

Not verified

- No national figure for the share of OPD patients who walk in versus book online, or for no-show rates, was found.
- The current per-SMS OTP price from MSG91 and Practo Ray's current price are not published. Both need quotes.
- The text of ASCI's healthcare guidelines could not be opened, because ascionline.in did not load. ASCI points come from secondary coverage.
- ABHA counts above 90 crore (for example 93.95 crore in July 2026) were seen only in search snippets.

Sources

The pages below were opened between 7 and 8 November 2026.

- Practo — About
- Apollo 24|7 homepage
- Google Play listings: Practo, Apollo 24|7, Max MyHealth, myFortis, Manipal Hospitals, NH Care, MediBuddy, Tata 1mg, Eka Care, Lybrate, Bajaj Finserv Health, eSanjeevani, ABHA, Zocdoc, NHS App
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